MIL-C-50790/1A(WC)
19 January 1977
SUPERSEDING
MIL-C-50790/1(WC)
13 June 1974

MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY: 8438903

(LOGIC, DIGITAL)

This specification is approved for use by the Rock Island Arsenal, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the circuit card assembly described herein shall consist of this document and the latest issue of specification MIL-C-50790.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for one type of circuit card assembly. The requirements of this specification together with the general requirements of MIL-C-50790 comprise the complete requirements for the Digital Logic Board.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-M-38510/1 - Microcircuits, Digital, TTL, NAND Gates, Monolithic Silicon

MIL-C-50790 - Circuit Card Assemblies, Test Set, AN/MWM-3 General Specification for

DRAWINGS

Rock Island Arsenal

8438903 - No. 1 Digital Logic Circuit, Board Assembly

FSC 4931

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PUBLICATIONS

Rock Island Arsenal
F8438903 - Packaging Data Sheet-No. 1 Digital Logic Circuit, Board Assembly

(Copies of specifications, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication. The assembly shall be manufactured in accordance with Drawing F8438903.

3.2 General specifications. The assembly shall meet the requirements of MIL-C-50790.

3.3 Environmental service conditions. As specified in MIL-C-50790

3.4 Electrical. Each conductor shall be electrically continuous as specified on the schematic diagram of the circuit card assembly.

3.5 Performance. When supplied with the inputs specified, the Digital Logic Circuit shall yield the outputs specified in paragraphs 3.5.1 through 3.5.16.

3.5.1 When pin P is connected to pin u, and ±5.0 + 0.5 volts direct current (vdc) is connected between pins m (+) and A (+) and pin P (-), and +26.0 ± 4.0 vdc is connected between pins e (+) and P (-), and +5.0 + 0.5 vdc is connected (through 1000 ohm, 1/4 watt, 5% resistors) between pins C (+), J (+), F (+), D (+), B (+) and pin P (-), the output voltage between pins J (+) and P (-) shall be + 4.5 ± 1.0 vdc.

3.5.2 With the inputs as in 3.5.1, when +26.0 + 4.0 vdc is connected sequentially between pins U (+), w (+), x (+) and P (-), the output between pins J (+) and P (-) shall be + 0.30 ± 0.25 vdc.

3.5.3 With the inputs as in 3.5.1, when +26.0 ± 4.0 vdc is connected between pins r (+) and P (-), the output between pins Y (+) and P (-) shall be + 4.5 ± 1.0 vdc.

3.5.4 With the inputs as in 3.5.1, when +8.8 ± 0.2 vdc is connected between r (+) and P (-); and +26.0 + 4.0 vdc is connected between V (+) and P (-) sequentially, the output between pins Y (+) and P (-) shall be + 4.5 ± 1.0 vdc.

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3.5.5 With the inputs as in 3.5.1, when +26.0 ± 4.0 vdc is temporarily connected between pins N (+) and P (-) and then K (+) and P (-), the output between pins V (+) and P (-) shall be a positive going pulse with a 3.5 ± 0.1 second duration and an amplitude of 4.5 ± 1.0 volts. This circuit shall be capable of being RESET to obtain the output pulse by temporarily replacing the plus lead of the + 5.0 ± 0.5 vdc at pin A (+) with ground.

3.5.6 With the inputs as in 3.5.1, when the RESET operation of 3.5.5 is performed, the output between pins D (+) and P (-) shall be +4.0 ± 1.0 vdc. When +5.0 ± 0.5 vdc is connected, in order, between pins E, h, b, N and pin P (-), the output between pins D (+) and P (-) shall be +0.30 ± 0.25 vdc when, and only when, all of the inputs have been connected in order.

3.5.7 With the inputs as in 3.5.1, when the RESET operation of 3.5.5 is performed, the output between pins D (+) and P (-) shall be +4.5 ± 1.0 vdc. When + 5.0 ± 0.5 vdc is connected, in order, between pins E, h, b, K and pin P (-), the output between pins D (+) and P (-) shall be + 0.30 ± 0.25 vdc when, and only when, all of the inputs have been connected in order.

3.5.8 With the inputs as in 3.5.1, when the RESET operation of 3.5.5 is performed, and +26.0 ± 4.0 vdc is connected between pins r (+) and pin P (-), the output between pins s (+) and P (-) shall be + 0.30 ± 0.25 vdc. The output between pins F, y and P (-) shall be + 4.5 ± 1.0 vdc. When + 8.0 ± 5% vdc is connected sequentially between pins r, y, and P (-), the output between pins F (+) and P (-) shall be + 0.30 ± 0.25 vdc.

3.5.9 With the inputs as in 3.5.1, when the RESET operation of 3.5.5 is performed, the output between pins s (+) and P (-) shall be +0.30 ± 0.25 vdc. When +26.0 ± 4.0 vdc is temporarily connected between pins l (+) and P (-), the output between pins s (+) and P (-) shall be +4.5 ± 1.0 vdc. When the RESET operation of 3.5.5 is again performed, and +26.0 ± 4.0 vdc is connected between pins c (+) and P (-), the output between pins s (+) and P (-) shall remain at +0.30 ± 0.25 vdc.

3.5.10 With the inputs as in 3.5.1, when the RESET operation of 3.5.5 is performed, the output between pins y (+) and P (-) shall be +4.5 ± 1.0 vdc. When +26.0 ± 4.0 vdc is temporarily connected between pins l (+) and P (-), the output between y (+) and P (-) shall be +0.30 ± 0.25 vdc. When the RESET operation of 3.5.5 is again performed, and +26.0 ± 4.0 vdc is connected between pins c (+) and P (-), the output between pins y (+) and P (-) shall remain at +4.0 ± 1.0 vdc.

3.5.11 With the inputs as in 3.5.1, when +27.50 ± 0.25 vdc is connected to pins w (+) and P (-), the output between pins M (+) and P (-) shall be +3.40 ± 0.05 vdc. This output shall be adjustable by operation of potentiometer R12.

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hardware which is used during servicing, accessing or installing test equip

3.5.12 With the inputs as in 3.5.1, when + 27.50 ± 0.25 vdc is connected between pins U (+) and pin P (-), the output between pins j (+) and P (-) shall be + 4.75 ± 0.50 vdc.

3.5.13 With the inputs as in 3.5.1, when the RESET operation of 3.5.5 is performed, the output between pins C (+) and P(-) shall be +4.5 ± 1.0 vdc. When ± 26.0 ± 4.0 vdc is connected between pins U (+) and P (-), and then +5.0 ± 0.5 vdc is connected between pins n (+) and P (-), and then +26.0 ± 4.0 vdc is connected between pins r (+) and P (-) in that order, the output between pins C (+) and P (-) shall be +0.30 ± 0.25 vdc only after all of the above inputs have been energized.

3.5.14 With the inputs as in 3.5.1, when the RESET operation of 3.5.5 is performed, the output between pins C (+) and P (-) shall be +4.5 ± 1.0 vdc. When + 26.0 ± 4.0 vdc is connected between pins S, w, z and pin P (-), and then +5.0 ± 0.5 vdc is connected between pins n (+) and P (-), and then +26.0 ±4.0 vdc is connected between pins r (+) and P (-) in that order, the output between pins C (+) and P (-) shall be +0.30 ± 0.25 vdc only after all of the above inputs have been energized.

3.5.15 With the inputs as in 3.5.1, when the RESET operation of 3.5.5 is performed, the output between pins B (+) and P (-) shall be + 4.5 + 1.0 vdc. When +26.0 ± 4.0 vdc is connected temporarily between pins N (+) and P (-), the output between pins B (+) and P (-) shall remain at +4.5 ± 1.0 vdc for 3.5 ± 0.5 seconds and then change to +0.30 ± 0.25 vdc. When the RESET operation of 3.5.5 is again performed and +26.0 ± 4.0 vdc is connected temporarily between pins K (+) and P (-), the output between pins B (+) and P (-) shall remain at +4.0 ± 1.0 vdc for 3.5 ± 0.5 seconds and then change to + 0.30 ± 0.25 vdc.

3.5.16 When +27.50 + 0.25 vdc is connected between pins z and P (-), the output between pins j (+) and P (-) shall be +4.75 ± 0.50 vdc.

3.6 Potentiometer adjustment. All potentiometers shall have a minimum of two turns remaining in both directions, after adjustment to the required setting.

4. QUALITY ASSURANCE PROVISIONS

4.1 General. The quality assurance provisions of MIL-C-50790 shall apply to the assembly covered by this detail specification.

4.2 First article inspection. As specified in MIL-C-50790.

4.3 Inspection provisions. As specified in MIL-C-50790.

4.4 Special sampling. As specified in MIL-C-50790.

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4.5 Inspection equipment. As specified in MIL-C-50790 and 4.5.3.

4.5.1 Government-furnished inspection equipment. As specified in MIL-C-50790.

4.5.2 Contractor-furnished inspection equipment. As specified in MIL-C-50790.

4.5.3 Test equipment. In conjunction with 4.5, the standard test equipment listed in Table I shall be utilized in performing the applicable tests of 4.6.

TABLE I - TEST EQUIPMENT LIST

ITEM NUMBER	DESCRIPTION
1. Oscilloscope	Dual Trace Tektronix Model 454 or equivalent
2. Digital Multimeter	Cimeron Model E9500B-1400 or equivalent
3. Multimeter	Western Reserve Model 300M-A or equivalent
4. D. C. Power Supply	Mid-Eastern Industries Inc. Series HW60-1.5 or equivalent

4.6 Test methods and procedures.

4.6.1 Environmental. As specified in MIL-C-50790.

4.6.2 Electrical. Continuity shall be inspected by standard measuring equipment (SME) and visual examination for compliance with 3.4.

4.6.3 Performance. The selected Logic, Digital circuit card shall be tested in accordance with 4.6.3.1 through 4.6.3.16.

4.6.3.1 Test set-up. The circuit card assembly shall be tested with the inputs and loads as shown in Figure I, unless otherwise specified (see 6.2).

4.6.3.2 Logic levels. The "0" level shall be between + 0.05 and + 0.55 vdc. The "1" level shall be between + 3.5 and + 5.5 vdc.

Note: After each test remove input(s) from circuit.

4.6.3.3 With the input levels as listed in Table I, a corresponding output level shall be obtained at Pin J.

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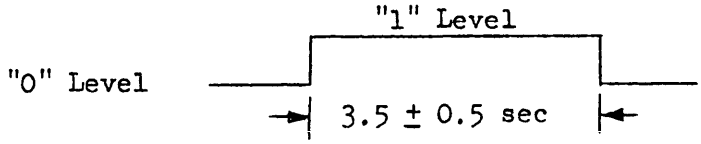
TABLE I

INPUT PIN LEVELS			OUTPUT LEVEL
U	W	X	Pin J
GND	GND	GND	"1"
+26 vdc	GND	GND	"0"
GND	+26 vdc	GND	"0"
GND	GND	+26 vdc	"0"

4.6.3.4 With GND applied to pin r, the output at pin Y shall be a "0" level. Remove GND from pin r and apply +26.0 ± 1.0 vdc. The output at pin Y shall be a "1" level.

4.6.3.5 Reset circuit by momentarily placing pin A to ground (see Figure 1). The output at pin y shall be a "1" level. Momentarily apply + 8.8 ± 0.2 vdc to pin r. The level at pin y shall be a "0" level. Reset circuit (pin y goes to "1" level) and apply + 26.0 ± 1.0 vdc to pin v. The voltage at pin y shall be a "0" level.

4.6.3.6 With + 26.0 ± 1.0 vdc momentarily applied to pin K, the output pulse at pin V shall be as shown below:



With + 26.0 ± 1.0 vdc momentarily applied at pin N, the output pulse at pin V shall be as shown above.

4.6.3.7 Reset circuit and apply GND to pins E, h, b and N. The output at pin D shall be a "1" level. Remove the above grounds and apply + 5.0 ± 0.5 vdc to pins E, h and b. The voltage at pin D shall remain at a "1" level. Apply + 26.0 ± 1.0 vdc to pin N. The voltage at pin D shall go to a "0" level.

4.6.3.8 Repeat 4.6.3.7, except pin K shall be substituted for pin N.

4.6.3.9 With the circuit reset, the output at pin s shall be a "0" level and at pins F and y a "1" level. Apply + 26.0 ± 1.0 vdc to pin r. The above levels shall remain the same. Remove + 26.0 ± 1.0 vdc from pin r and apply + 8.8 ± 0.2 vdc to pin r. The output at pin s shall be a "1" level and at pins F and y a "0" level.

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4.6.3.10 With the circuit reset, the output at pin s shall be a "0" level. Momentarily apply + 26.0 ± 1.0 vdc to pin c. The voltage at pin s shall remain a "0" level. Apply + 26.0 ± 1.0 vdc to pin 1. The voltage at pin s shall change to a "1" level.

4.6.3.11 With + 27.50 ± 0.25 vdc applied to pin w, the output at pin M shall be adjusted to + 3.40 ± 0.05 vdc by potentiometer R12.

4.6.3.12 With ground applied to pin U, the output at pin j shall be a "0" level. Remove ground from pin U and apply +27.50 ± 0.25 vdc. The output at pin j shall be 4.75 ± 0.50 vdc.

4.6.3.13 With the circuit reset, the output at pin C shall be a "1" level. Apply + 26.0 ± 1.0 vdc to pin U and + 5.0 ± 0.5 vdc to pin n. The output at pin C shall remain at a "1" level. Apply + 26.0 ± 1.0 vdc to pin r. The voltage at pin C shall be a "0" level.

4.6.3.14 With the circuit reset, the output at pin C shall be a "1" level. Apply + 26.0 ± 1.0 vdc to pins S, w and z and + 5.0 ± 0.5 vdc to pin n. The output at pin C shall remain at a "1" level. Apply + 26.0 ± 1.0 vdc to pin r. The voltage at pin C shall be a "0" level.

4.6.3.15 With the circuit reset, the voltage on pin B shall be a "1" level. Momentarily apply + 26.0 ± 1.0 vdc to pin N. The voltage at pin B shall remain at a "1" level for 3.5 seconds after voltage is removed from pin N and then change to a "0" level.

4.6.3.15.1 Repeat 4.6.3.15 except that pin K shall be substituted for pin N.

4.6.3.16 With ground applied to pin z, the output at pin j shall be a "0" level. Remove ground from pin z and apply + 27.50 ± 0.25 vdc. The output at pin j shall be + 4.75 ± 0.50 vdc.

4.7 Potentiometer adjustment. After adjustment to the required settings, verify that all potentiometers have a minimum of two turns remaining in both directions.

5. PACKAGING

5.1 Preservation, packing and marking. Preservation, packing and marking shall be in accordance with Packaging Data Sheet P8438903, for the level of protection specified (see 6.2).

6. NOTES

6.1 Intended use. The Digital Logic Circuit Board is used in the Test Set, AN/MSM-3, CMTS for the Vulcan Air Defense Weapon System. The Digital Logic Circuit Board provides Digital Logic which determines if the system relays have performed their function.

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. List of drawings, specifications and publications pertinent to the circuit card assembly, showing applicable revision dates.
c. Inputs and loads for testing, if other than as specified (see 4.6.3.1).
d. Selection of applicable level of preservation, packing and marking (see 5.1).
e. Applicable stock number.
f. Requirements for first article submission.

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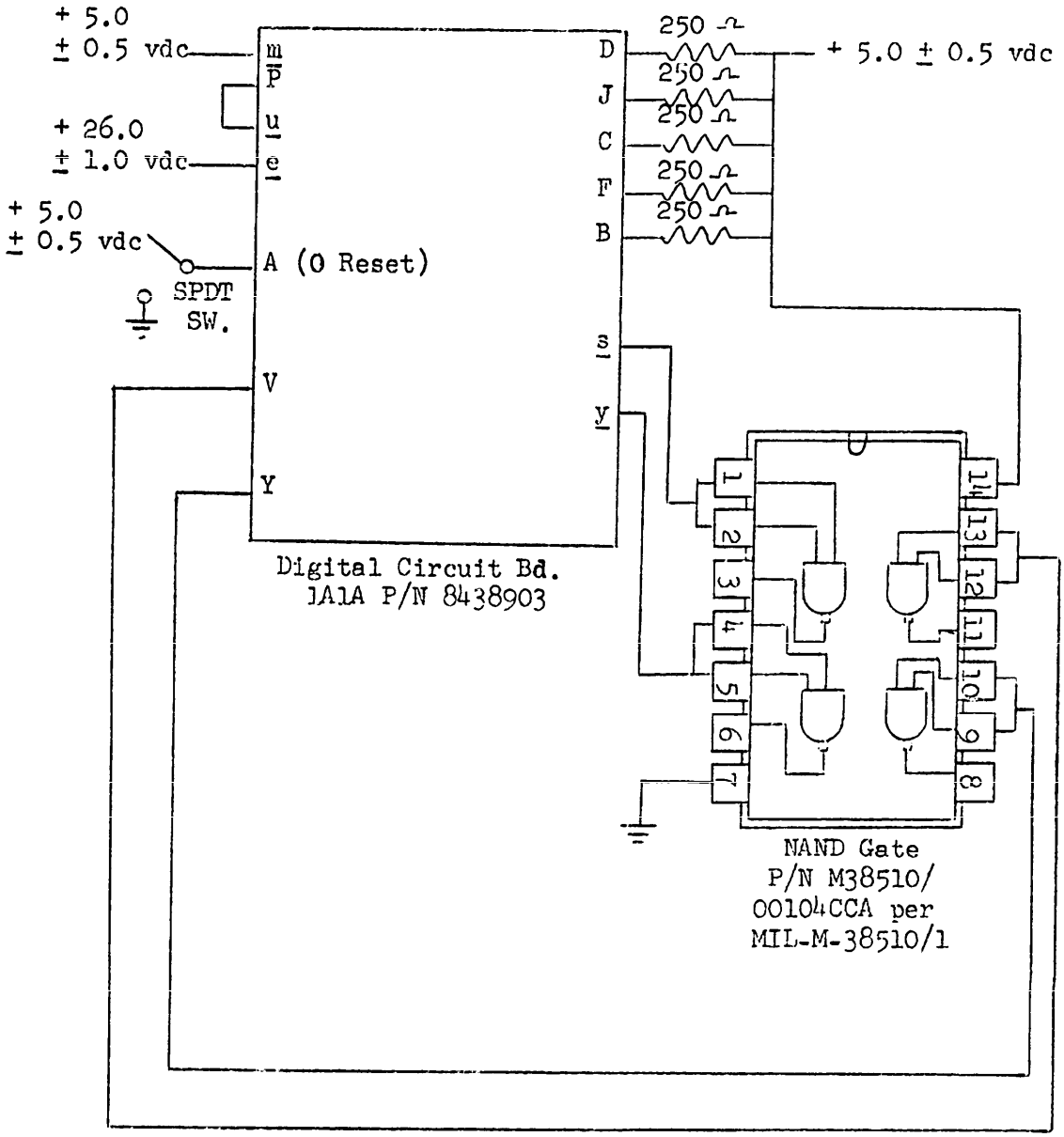


FIGURE 1. Test setup schematic.

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